

DIGITAL BREAST TOMOSYNTHESIS VERSUS BREAST MRI: SENSITIVITY AND SPECIFICITY FOR BREAST CANCER DETECTION IN WOMEN WITH DENSE BREASTS - A SYSTEMATIC REVIEW

Keywords: Breast MRI • Digital Breast Tomosynthesis (DBT) • Dense Breast • Breast Cancer Screening • Sensitivity and Specificity

Irfan Hadi Setyowijoyo¹, Vininta Fazharyasti^{3,5}, Terawan Agus Putranto^{4,5}, Nataliandra Natasha Maarifat^{2,5}

¹Radiology Resident, Dept. of Radiology, Faculty of Medicine, Universitas Pembangunan Nasional Veteran, Jakarta, Indonesia • ²Women's Imaging Radiologist, ³Thoracic Radiologist, ⁴Interventional Radiologist, Dept. of Radiology, Gatot Soebroto Army Hospital, Jakarta, Indonesia • ⁵Faculty of Medicine, Dept. of Radiology, Universitas Pembangunan Nasional Veteran Jakarta, Indonesia



BACKGROUND

- In dense breasts (BI-RADS C-D) cancer and normal parenchyma are both radiopaque and superimpose on 2D mammography.
- Mammographic sensitivity falls to 30–50%, with more false negatives and interval cancers.
- DBT reduces tissue superimposition; Breast MRI detects neoangiogenesis and is largely density-independent.
- Their comparative accuracy remains contested.

OBJECTIVES

- Primary: to compare the sensitivity and specificity of DBT versus Breast MRI in women with dense breasts.
- Secondary: to stratify findings by clinical setting and BI-RADS density category.
- Outcomes: sensitivity and specificity only.

METHODS

- PRISMA 2020 systematic review; PubMed, Semantic Scholar and Google Scholar.
- Women >18 y with BI-RADS C-D density; index test DBT, comparator Breast MRI (full DCE or abbreviated).
- Reference standard: histopathology or ≥1 year of negative imaging and clinical follow-up.
- Excluded: reviews, editorials, cost models, series <20 patients, non-English reports.
- QUADAS-2 risk of bias; narrative synthesis (SWiM) by clinical setting. Patient selection was high risk in 6 of 14 reports; index test (12/14) and reference standard (11/14) were mostly low risk

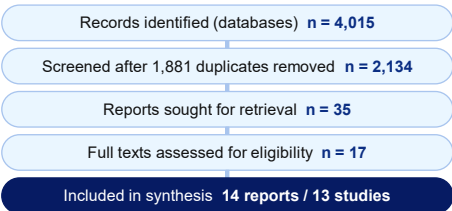


Figure 1. PRISMA 2020 study selection flow.

RESULTS

14 reports / 13 studies included (n = 30–4,091 women). QUADAS-2: patient selection low risk in 8/14 reports; index test and ref. standard mostly low risk.

Table 1. Sensitivity and specificity in all 14 included reports (%)

Study	Setting	DBT Sens.	MRI Sens.	DBT Spec.	MRI Spec.
Aloufi 2025	Diagnostic	89.5	94.7	52.8	55.6
Azzam 2020	Diagnostic	86.0	–	81.0	–
Barakat 2022 ‡	Surveillance	88.9	88.9	95.2	90.5
Chikarmane 2021	Surveillance	96.4	–	92.3	–
Comstock 2020	Screening (prevalence)	39.1	95.7	97.4	86.7
Förnvik 2018	Diagnostic	80.6	97.1	N/R	N/R
Kuhl 2023	Screening (incidence)	50.0	60.0	98.5	93.3
Nadler 2017	Surveillance	–	100.0	–	80.5
Narayan 2016 §	Screening (high risk)	–	100.0	–	73.0
Petrović 2025	Diagnostic	88.9	95.1	56.0	78.7
S. Aragon 2021	Screening	54.0	–	88.0	–
Savaridas 2026	Diagnostic	94.2	98.5	60.0	54.6
Tang 2017 †	Diagnostic	92.1	94.7	73.0	82.3
Yi 2018 ¶	Diagnostic	88.6	–	95.6	–

Pink rows = both modalities evaluated in the same women. † DBT + mammography vs MRI + mammography. ‡ DBT + ultrasound vs Breast MRI. § comparator was mammography (0.0% / 80.6%). ¶ DBT + FFDM overall; BI-RADS D: 63.6% / 84.8%. – = not evaluated. N/R = not reported.

Figure 2. Sensitivity of DBT versus Breast MRI in the 8 paired reports

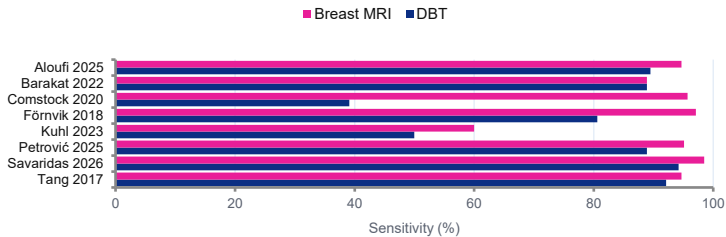
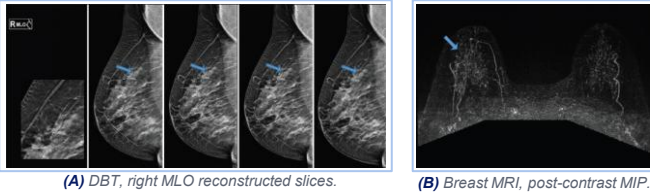


Figure 3. Representative imaging appearance in a dense breast



Key numerical findings

- MRI sensitivity exceeded DBT in every paired study (88.9–100.0% vs 39.1–96.4%); largest gap prevalence round - 95.7% vs 39.1% in the same 1,444 women.
- Specificity favoured DBT in screening (97.4–98.5% vs 86.7–93.3%) but reversed in diagnostic cohorts (MRI 78.7–90.5%; DBT 52.8–81.0%).
- Density-stratified data from one study: DBT 95.2% (BI-RADS A-C) vs 63.6% (BI-RADS D); both fell in the incidence round (MRI 60.0%; DBT 50.0%).

CONCLUSION

1. Main finding

Breast MRI is consistently more sensitive than DBT in dense breasts, reaching 95.7% versus 39.1% in the same screening cohort.

2. Clinical meaning

DBT remains the more specific screening test (97.4 - 98.5%); accuracy is driven by clinical setting and pretest probability, not by modality alone.

3. Next step

Integrate both in a risk-stratified pathway: DBT as the primary examination, supplemental MRI for extremely dense or high-risk breasts.

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- The review protocol was not prospectively registered in PROSPERO.